

Living Off The Pipeline: Defensive Research, Weaponized (SmokedMeat / Brisket) (Tool)

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Content

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This reference is a source-code repository. The archive preserves its documentation at an exact commit; the code itself stays in a private mirror and is never checked out, built or run.

- Repository: <<https://github.com/boostsecurityio/smokedmeat>>
- Commit: 3b2873dbecac0d7c5a66d32bbc5de4865342f8da
- Documents preserved: 16

CHANGELOG.md

Blob 2193f5a3f0dc, 5922 bytes, at commit 3b2873dbecac.

Changelog

All notable changes to SmokedMeat are documented in this file.

The format follows [Keep a Changelog](#), and releases use [Semantic Versioning](#).

[Unreleased]

Added

- Canonical domain language in `CONTEXT.md` and an ADR for Kitchen schema compatibility.
- Agent-specific guidance for engineering, TUI work, persistence, issue tracking, and triage.
- TROOPERS slide-deck images for project documentation.

Changed

- Replaced the repository roadmap files with GitHub Issues and the private SmokedMeat Roadmap Project as planning sources of truth.
- Reduced `AGENTS.md` to durable constraints and progressive links to task-specific guidance.
- Rebuilt this changelog from the published GitHub release history.
- Pinned release-backed quickstart to v0.3.1.

[0.3.1] - 2026-06-23

Added

- Trusted self-hosted runner harvest results now create normal cloud shell sessions.
- Resident cloud credentials include their expiry window when available.

Changed

- Docker shell containers mount host user and group databases for reliable UID and GID resolution on macOS and Linux.
- Linux lint coverage includes the resident-harvest path.

Fixed

- Release-backed Docker SSH shells no longer fail when the container cannot resolve the host UID.
- SSH private keys are canonicalized before containerized clients load them.
- Resident harvest tolerates short-lived process races.
- Trusted runner cloud tokens remain available after resident harvest.

Kitchen schema remains compatible with v0.3.0. No purge is required.

[0.3.0] - 2026-06-05

Added

- Authenticated browser-based GitHub source and repository browsing through Kitchen.
- Workflow source inspection from Counter with source-aware findings.
- GitHub Actions Auditor mode for browser-native workflow review.
- Custom Poutine rule packs with configuration, validation, mappings, and Counter rule summaries.

Changed

- Graph nodes link into the browser source viewer.
- Repository browsing includes richer navigation and control-plane context.

- Cache-poisoning results preserve stronger runtime correlation.

Fixed

- Gump recovers runtime context on slim runners without the usual process environment path.
- Quickstart state works correctly for same-host multi-user installations.
- SSH shell launch resolves its entrypoint from the current environment.
- Imported Pantry findings use their normalized exploit class.

Kitchen schema advances from 2.4 to 2.5 and remains on major version 2. No purge is required from v0.2.0.

[0.2.0] - 2026-05-05

Added

- Self-hosted runners as first-class attack graph and Counter targets.
- Resident Brisket footholds with later-job observation and automatic harvest.
- Workflow dispatch targeting with input review, triggering, and correlated waiting state.
- Counter release update checks.

Changed

- Waiting, callback, loot, Pantry, graph, tree, and omnibox views carry richer execution context.
- Re-analysis replaces stale repository findings before importing fresh results.
- Open-target actions identify the actual workflow, run, issue, pull request, or comment.

Fixed

- Public authentication request bodies are size-limited.
- Authentication challenges have per-IP rate limiting and bounded pending state.
- Gitleaks findings are attributed to the correct repository.
- Self-hosted runner signals remain analyze-only until an honest exploit path exists.

Kitchen schema remains on major version 2. No purge is required from v0.1.2.

[0.1.2] - 2026-04-24

Added

- Per-source injection variants are persisted with stable discriminators.

Changed

- Automatic LOTP delivery is limited to payload families with working generators and delivery paths.

- Detected but unsupported LOTP findings remain available as analyze-only findings.
- **Breaking:** Kitchen schema advances from 1.0 to 2.0. Existing v0.1.0 and v0.1.1 volumes must be purged before upgrade.

Fixed

- Stager metadata is accessed through locked snapshots instead of shared mutable pointers.
- Pip, Yarn, Cargo, and Make prefer callback-bearing payload variants.
- Setup action aliases resolve to honest LOTP generators.

[0.1.1] - 2026-04-17

Added

- Bash injection context analysis for command, argument, quoting, and heredoc positions.
- Branch-name payload delivery for `github.head_ref` findings.
- LOTP wizard previews built on Ultraviolet compositing.

Fixed

- Public quickstart resolves downloads against the pinned release path.

Kitchen schema is unchanged from v0.1.0. No purge is required.

[0.1.0] - 2026-04-14

First public release. GitHub Actions is the supported analysis, delivery, exploitation, and pivot platform. Other CI providers are detected for runner classification only.

Added

- Counter operator client, Kitchen teamserver, and Brisket implant flow.
- GitHub Actions analysis for injection paths, dangerous triggers, and unsafe checkout patterns.
- Payload delivery through pull requests, issues, comments, LOTP, and workflow dispatch.
- Runner post-exploitation, secret extraction, token enumeration, and cloud pivots.
- Persistent Pantry attack graph with a live browser view.
- Release-backed and development quickstart flows.
- Whooli playground, tutorial, feature reference, and public deployment documentation.

[Unreleased]: <https://github.com/boostsecurityio/smokedmeat/compare/v0.3.1...HEAD> [0.3.1]:

<https://github.com/boostsecurityio/smokedmeat/compare/v0.3.0...v0.3.1> [0.3.0]:

<https://github.com/boostsecurityio/smokedmeat/compare/v0.2.0...v0.3.0> [0.2.0]:

<https://github.com/boostsecurityio/smokedmeat/compare/v0.1.2...v0.2.0> [0.1.2]:

<https://github.com/boostsecurityio/smokedmeat/compare/v0.1.1...v0.1.2> [0.1.1]:

<https://github.com/boostsecurityio/smokedmeat/compare/v0.1.0...v0.1.1> [0.1.0]:
<https://github.com/boostsecurityio/smokedmeat/releases/tag/v0.1.0>

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Blob `e3548f7f3ee1`, 1557 bytes, at commit `3b2873dbecac`.

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```
I have read the CLA Document and I hereby sign the CLA
```

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Blob `be3f7b28e564`, 34523 bytes, at commit `3b2873dbecac`.

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README.md

Blob `b5ffb868888d`, 14109 bytes, at commit `3b2873dbecac`.

SmokedMeat

[[image: License: AGPL v3]](<https://www.gnu.org/licenses/agpl-3.0>)

CI/CD Red Team Framework

Like Metasploit, but for CI/CD pipelines.

From the makers of the [poutine](#) Build Pipeline SAST scanner at [BoostSecurity Labs](#).

[\[image: SmokedMeat quickstart demo\]](#)

Warning: This tool is for authorized security testing only. SmokedMeat exists because CI/CD pipeline threats are deeply underestimated. Traditional security training rarely covers supply chain attacks, leaving defenders unprepared for techniques that adversaries actively exploit in the wild. We built this to give security teams the ability to learn, practice, and validate defenses against advanced CI/CD attack techniques through realistic red team exercises. **Only use against systems you own or have explicit written permission to test.**

What is SmokedMeat?

SmokedMeat is a post-exploitation framework for CI/CD pipelines. Point it at a GitHub organization, let it find vulnerable workflows, deploy an implant to a compromised runner, then pivot through cloud providers, extract secrets, and map the blast radius - all from a terminal UI.

What it does:

1. **Analyze** - Scan an org's GitHub Actions workflows for injection vulnerabilities, dangerous triggers, and unsafe checkout patterns (powered by [poutine](#))
2. **Exploit** - Deploy a stager via PR, issue, comment, or workflow dispatch. When the vulnerable workflow runs, it downloads and executes the implant on the CI runner.
3. **Post-exploit** - Extract secrets from runner memory, enumerate GitHub token permissions, scan for private keys, and collect loot

4. **Pivot** - Use captured credentials to move laterally: discover private repos, mint GitHub App tokens, exchange OIDC tokens for AWS/GCP/Azure access, probe SSH deploy keys

Philosophy: Bold and noisy. This isn't an EDR evasion tool. It's a demonstration framework that shows how deep a CI/CD compromise goes before anything triggers an alert.

Who is it for:

- Red teams validating CI/CD security posture in enterprise environments
- Pentesters demonstrating supply chain attack paths to stakeholders
- Security engineers testing detection and response for pipeline attacks
- Researchers developing new CI/CD exploitation techniques
- Bug bounty hunters exploring supply chain attack surface

Quick Start

To try SmokedMeat for the first time, install Docker and `make`. Go is not required.

```
git clone https://github.com/boostsecurityio/smokedmeat.git
cd smokedmeat
make quickstart
```

`make quickstart` is the recommended first run. It starts the stable release quickstart stack locally and launches the operator TUI (`Counter`) against the local C2 teamserver (`Kitchen`).

Recommended first run:

- Target: `whooli`
- Token: classic PAT with `public_repo`

Prefer a classic PAT. Fine-grained PATs can be too restrictive and may block testing public targets in other orgs, including `whooli`.

`whooli` is SmokedMeat's deliberately vulnerable CI/CD attack playground. It is the recommended first target for the public path.

The setup wizard walks you through:

1. **GitHub PAT** - Enter your token. For private repos, a classic PAT will usually need `repo`.
2. **Target** - Enter `whooli` or your own org/repo
3. **Analysis** - Scans workflows for vulnerabilities and presents exploitable findings

For the full challenge flow, see the `whooli` guide or go straight to the [whooli GitHub org](#).

When you are done:

```
make quickstart-down    # Stop containers
make quickstart-purge   # Stop and delete all data
```

If you want to work from source instead, see Development.

Development

If you are contributing or iterating on the source tree locally, install Go 1.26+ and use the dev quickstart:

```
make dev-quickstart
```

`make dev-quickstart` builds the local `smokedmeat-cloud-shell` image, starts `cloudflared`, `nats`, and the C2 teamserver (`Kitchen`), then launches the operator TUI from source.

If you want the infrastructure first and the operator TUI later:

```
make dev-quickstart-up
make dev-quickstart-counter
```

When you are done:

```
make dev-quickstart-down # Stop containers
make dev-quickstart-purge # Stop and delete all data
```

More deployment modes and local development details are in `docs/deployment.md`.

Version Check

Counter checks for newer SmokedMeat releases at startup. It can be disabled by setting the `SMOKEDMEAT_DISABLE_VERSION_CHECK` environment variable.

Core Components

Standard term	SmokedMeat name	Description
Operator TUI	Counter	Terminal interface for analysis, payload delivery, and post-exploitation workflow.
C2 teamserver	Kitchen	API and WebSocket server for operator sessions, stagers, callbacks, and graph state.
Implant	Brisket	Agent delivered to compromised CI runners for beaconing, command execution, and pivoting.
Browser graph view	Browser View	Live attack graph served by the C2 teamserver at <code>/graph</code> .

Deployment Modes

Mode	Use it when	Entry point
Quickstart	Fastest first run on the pinned release	<code>make quickstart</code>
Dev Quickstart	Working on the source tree locally	<code>make dev-quickstart</code>
Hosted Teamserver	Running a real engagement with a stable domain	<code>docs/deployment.md</code>

Hosted Teamserver runs the C2 teamserver on a dedicated host and the operator TUI natively on each operator workstation.

Architecture

At a high level, the operator TUI (`Counter`) talks to the C2 teamserver (`Kitchen`), which manages implants (`Brisket`) running on compromised CI runners and serves the live attack graph.

SSH AGENT
(Auth)

THE COUNTER THE KITCHEN
(Operator) WebSocket (Teamserver)
Bubbletea
TUI Events/Graph
Database

BROWSER
Graph View WebSocket
Visualizer
Live Updates

Creates PR Stager fetches implant binary
Implant HTTP Beacon/Commands

GITHUB.COM

Malicious PR triggers GitHub Actions Runner
(Vulnerable
Workflow)

Stager THE BRISKET
(Implant)

Features

Full details in docs/FEATURES.md.

Category	Capabilities
Reconnaissance	Auto-detect 6 CI platforms (GitHub Actions, GitLab CI, Azure DevOps, CircleCI, Jenkins, Bitbucket). Classify secrets, probe OIDC availability, gather runner

Category	Capabilities
	metadata.
Secret Extraction	Scan Runner.Worker process memory via <code>/proc</code> to recover unmasked <code>secrets.*</code> , <code>vars.*</code> , and <code>GITHUB_TOKEN</code> permission maps that GitHub hides from logs.
Vulnerability Analysis	Embedded <code>poutine</code> SAST for injection vulnerabilities, dangerous triggers, and workflow <code>if:</code> gate classification. Gitleaks deep scan for private keys and PATs in git history.
Delivery	5 automated methods: PR, issue, comment, LOTP, workflow dispatch - plus copy-only and manual. Draft PR support, auto-close on callback, server-side dispatch preflight.
Injection Payloads	Context-aware payload generation for 8 injection vectors (branch name, PR title/body, commit message, issue title/body, github-script, bash run) with constraint-aware techniques.
LOTP	Living Off The Pipeline catalog: 15 build tools (npm, pip, cargo, make, docker, gradle, maven, and more) with config-file payloads for code execution during install/build/test.
Cache Poisoning	Writer/victim classification, exact cache key prediction, archive staging via the Actions Cache API. Wizard-driven flow with implant arming.
Token Enumeration	Probe GitHub tokens against API endpoints to enumerate 10 permission scopes, identify token type, and list accessible repos and orgs.
Cloud Pivots	OIDC token exchange for AWS (<code>sts:AssumeRoleWithWebIdentity</code>), GCP (Workload Identity Federation), Azure (AAD), and Kubernetes. Post-pivot resource enumeration.
Cloud Shell	Durable local sessions with <code>cloud shell</code> (pre-configured <code>gcloud/aws/az</code>), <code>cloud export</code> , and provider quick checks.
SSH Pivoting	Probe repos for SSH deploy key access (read/write), <code>ssh shell</code> with temporary agent, confirmed access persisted to graph.
GitHub Pivoting	<code>pivot github</code> for repo discovery, <code>pivot app</code> for GitHub App PEM-to-installation-token exchange. Discovered repos auto-queued for analysis.
Attack Graph	Persistent directed graph (BBolt) with <code>org/repo/workflow/job/vuln/token/cloud</code> nodes. Live Cytoscape.js browser visualization at <code>/graph</code> .
Operator TUI	Phase-aware workflow, 7-step setup wizard, attack tree navigation, exploit wizard, loot stash, omnibox search, tab completion, OSC 8 hyperlinks.
Teamserver	SSH or token auth, NATS JetStream message bus, GitHub API proxy (tokens stay server-side), auto-TLS via Caddy, operation history.

Technology Stack

Layer	Technology
Language	Go 1.26+
TUI Framework	Bubbletea v2 + Lipgloss v2
TUI Layout	Ultraviolet layout + ANSI-safe screen compositing
Message Bus	NATS JetStream
Attack Graph	hmdsefi/gograph
Graph Visualization	Cytoscape.js
Database	BBolt
CI/CD Scanner	poutine (embedded)
Secret Scanner	gitleaks (embedded, custom rules)
Runner Secret Extraction	gump (embedded, <code>/proc</code> memory scanning)
Cloud SDKs	AWS SDK v2, Google Cloud, Azure SDK for Go
Reverse Proxy	Caddy (auto-TLS)

Testing

```
make test      # Unit tests
make lint      # Linter
make e2e-smoke # Fast public exploit smoke path
make e2e-goat  # Full goat chain to the cloud flag
```

Prior Art

SmokedMeat builds on research from:

- [poutine](#) - Build Pipeline SAST scanner
- [LOTP](#) - Living Off The Pipeline techniques
- [Gato-X](#) - GitHub Actions enumeration
- [Nord-Stream](#) - CI/CD secret extraction
- [Sliver](#) - Go C2 architecture patterns
- [Mythic](#) - Collaborative workflow design

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Built for defenders who want to understand attacker techniques.

Blob 9537348fb6ce, 1004 bytes, at commit 3b2873dbecac.

Security Policy

Reporting a Vulnerability

We take security vulnerabilities seriously. If you discover a security issue in SmokedMeat, please report it privately using GitHub's Security Advisory feature:

1. Go to the Security tab of this repository
2. Click "Report a vulnerability"
3. Fill out the private security advisory form

Please do not report security vulnerabilities through public GitHub issues.

What to Include

When reporting a vulnerability, please include:

- Description of the vulnerability
- Steps to reproduce
- Potential impact
- Suggested fix (if any)

Response Timeline

- We will acknowledge receipt within 48 hours
- We will provide an initial assessment within 7 days
- We will work with you to understand and resolve the issue

Scope

This security policy applies to the SmokedMeat framework itself. For vulnerabilities in CI/CD pipelines discovered using SmokedMeat, please report those to the respective platform or organization owners.

Blob 4e23ed849b15, 11942 bytes, at commit 3b2873dbecac.

SmokedMeat Features

Detailed reference for every capability in the framework. For a high-level overview, see the README.

Reconnaissance

The Brisket implant auto-detects the CI platform it's running on and collects environment details, secrets, OIDC token availability, runner metadata, and network egress capabilities.

For the first public release, SmokedMeat is intentionally scoped to GitHub Actions. The non-GitHub detections below are useful for runner classification and recon notes, but the supported analysis, delivery, exploit, token-pivot, cache-poisoning, and cloud-pivot workflows are GitHub Actions only in v0.1.0 .

Platform	Detection	Public v0.1.0 Support
GitHub Actions	<code>GITHUB_ACTIONS=true</code>	Full support
GitLab CI	<code>GITLAB_CI=true</code>	Recon classification only
Azure DevOps	<code>TF_BUILD=True</code>	Recon classification only
CircleCI	<code>CIRCLECI=true</code>	Recon classification only
Jenkins	<code>JENKINS_URL</code>	Recon classification only
Bitbucket Pipelines	<code>BITBUCKET_BUILD_NUMBER</code>	Recon classification only

The `recon` command also classifies secrets found in environment variables by type (AWS keys, GitHub tokens, GCP service accounts, Slack tokens, database URLs, generic API keys, etc.) and probes for OIDC provider availability (AWS, GCP, Azure).

Runner Secret Extraction (gump)

On Linux runners, the `env` command goes beyond `os.Environ()` by scanning the GitHub Actions Runner.Worker process memory via `/proc/<pid>/mem` . This recovers the full unmasked values of `secrets.*` and `vars.*` that GitHub normally masks in logs, along with the `GITHUB_TOKEN` and its fine-grained permission map.

Vulnerability Analysis

SmokedMeat embeds `poutine` for static analysis of GitHub Actions workflows. The `analyze` command sends a target org or repo to the Kitchen, which clones and scans it for:

- **Injection vulnerabilities** - Untrusted input (`github.event.pull_request.title` , `github.event.issue.body` , etc.) flowing into `run:` blocks or `actions/github-script`
- **Dangerous triggers** - `pull_request_target` , `issue_comment` , `workflow_run` with unsafe checkout patterns
- **Workflow if: gate detection** - Classifies gate conditions (e.g., `contains(github.event.comment.body, '/deploy')`) into trigger types with extracted trigger strings

Results populate the attack tree: org > repo > workflow > job > vulnerability.

The `deep-analyze` command adds Gitleaks-based secret scanning (private keys, GitHub PATs, fine-grained PATs, PKCS#12 files) across the repository's git history.

Delivery Methods

The exploit wizard supports 5 automated delivery methods plus 2 manual fallbacks, selected based on the vulnerability type:

Method	How it works
Create PR	Fork the repo, push a branch with the stager payload, open a PR to trigger <code>pull_request_target</code> workflows. Supports draft PRs and auto-close on callback.
Create Issue	Open an issue with the payload in the title/body to trigger <code>issues</code> -triggered workflows. Supports auto-close on callback.
Add Comment	Comment on an existing issue or PR to trigger <code>issue_comment</code> -triggered workflows. Can target issues, existing PRs, or create a stub PR.
LOTP	Fork the repo, inject a Living Off The Pipeline payload into a build config file, and open a PR.
Trigger Dispatch	Send a <code>workflow_dispatch</code> event with the payload. Includes server-side preflight validation (workflow existence, required inputs).
Copy Only	Copy the stager payload to clipboard for manual delivery.
Manual Steps	Display step-by-step instructions.

Each deployment registers a stager URL on the Kitchen. When the stager executes in the CI runner, it downloads the Brisket binary and starts beaconing.

Injection Payloads (Rye)

The `inject` command generates context-aware injection payloads for 8 GitHub Actions injection vectors:

Context	Constraints	Techniques
<code>git_branch</code>	250 chars, no spaces/special chars	<code>\$IFS</code> -separated substitution
<code>pr_title</code>	256 chars, single line	Backtick/ <code>\$()</code> substitution, pipe/chain injection
<code>pr_body</code>	64KB, multiline	Newline injection, quote breaking
<code>commit_message</code>	72 chars first line	Substitution, chaining
<code>issue_title</code>	256 chars, single line	Same as <code>pr_title</code>
<code>issue_body</code>	64KB, multiline	Same as <code>pr_body</code>

Context	Constraints	Techniques
github_script	Unlimited, JavaScript	Template literal escape, child_process.execSync , process.mainModule sandbox bypass
bash_run	Unlimited, bash	All bash techniques

Living Off The Pipeline (LOTP)

The LOTP catalog covers 15 build tools that execute code during install, build, or test phases:

npm, Yarn, pip, Bundler, Cargo, Go (go generate), Make, Docker, ESLint, Prettier, Jest, Gradle, Maven, Composer, pre-commit, Husky

Each entry includes the config files that enable the technique, the commands that trigger it, and example payloads. The lotp delivery method injects these payloads into the appropriate config file via a forked PR.

Cache Poisoning

SmokedMeat can poison GitHub Actions caches to achieve persistence across workflow runs. The cache poisoning module:

- **Classifies writer eligibility** - Determines which vulnerabilities can write to the cache (requires a trigger that runs attacker-controlled code)
- **Collects victim candidates** - Identifies downstream workflows that restore from the same cache keys (actions/setup-go , actions/cache , etc.)
- **Computes exact cache keys** - Predicts the cache key using the same hash algorithm GitHub uses (Go version + go.sum hash, or explicit key patterns)
- **Stages replacement entries** - Builds a poisoned cache archive and uploads it via the Actions Cache API

The wizard walks the operator through selecting a writer vulnerability, choosing a victim workflow, and deploying the poisoned cache with an armed implant.

GitHub Token Enumeration

The token-test command probes the available GitHub token against API endpoints to enumerate its actual permissions: repo , read:user , user:email , user:follow , read:ssh_signing_key , read:gpg_key , read:org , gist , actions , read:packages . It also identifies the token type (classic PAT, fine-grained PAT, GitHub Actions token, OAuth, installation token) and lists accessible repositories and organizations.

OIDC Cloud Pivots

When the implant runs on a GitHub Actions runner with OIDC configured, oidc <provider> extracts a federated token and oidc pivot <provider> uses it to authenticate to the cloud provider:

Provider	Token Exchange	Post-Pivot Queries
AWS	sts:AssumeRoleWithWebIdentity	sts:GetCallerIdentity , S3 bucket listing, ECR repository listing
GCP	sts.googleapis.com + iamcredentials	Project listing, GCS bucket listing, caller identity
Azure	AAD token exchange	Subscription listing, resource group listing, storage accounts, ACR listing
Kubernetes	Direct OIDC token	Token extraction only

Cloud Post-Exploit (Counter)

After a successful OIDC pivot, the Counter provides:

- **Durable cloud sessions** - Credentials persist locally across TUI restarts
- **cloud shell** - Drop into a local shell with pre-configured cloud CLI credentials (`gcloud` , `aws` , `az`). For GCP, this bootstraps a local `gcloud` credential database with the OIDC access token.
- **cloud export** - Print the shell `export` commands for the active cloud session
- **Provider quick checks** - One-command enumeration (identity, buckets, projects, etc.) surfaced directly in the TUI

SSH Pivoting

When the implant recovers an SSH private key (deploy key or user key) from secrets or memory:

- **pivot ssh** - Test the key against the current target repo via `ssh -T git@github.com`
- **pivot ssh org:<owner>** - Probe all known repos in an org for SSH access
- **pivot ssh repo:<owner/repo>** - Probe a specific repo
- **ssh shell** - Drop into a local shell with the SSH key loaded in a temporary `ssh-agent` , with helper scripts for `git clone` via SSH
- **ssh status** - Show confirmed SSH access from the attack graph

Confirmed repo access (read-only vs read-write) is persisted into the Pantry graph labels.

GitHub Token Pivoting

- **pivot github [target]** - Use a captured PAT or `GITHUB_TOKEN` to list accessible repos, discover private repos, and find new attack surface. Discovered repos are automatically queued for analysis.
- **pivot app [app_id]** - Exchange a captured GitHub App private key (PEM) for a JWT, then mint an installation token. The token inherits the App's installation permissions.

Attack Graph (Pantry)

All discoveries feed into a persistent directed graph stored in BBolt:

- **Node types** - Organizations, repositories, workflows, jobs, vulnerabilities, tokens, cloud resources, OIDC providers
- **Edge types** - Contains, triggers, grants-access, pivots-to
- **Pivot rules** - Automated suggestions based on node types and captured credentials
- **Browser visualization** - Live Cytoscape.js graph at `/graph` with WebSocket updates as new nodes are discovered

Operator Interface (Counter)

The Counter TUI provides:

- **Phase-aware workflow** - Setup > Recon > Wizard > Waiting > Post-Exploit with context-sensitive commands at each phase
- **7-step setup wizard** - Kitchen URL, SSH key selection, operator name, key deployment, GitHub PAT, target org/repo, and initial analysis
- **Attack tree navigation** - Expandable org/repo/workflow/job/vuln hierarchy with keyboard navigation. Private repos are highlighted. Nodes link to GitHub via OSC 8 hyperlinks.
- **Exploit wizard** - 3-step flow: select vulnerability, choose delivery method, configure options (draft PR, auto-close, dwell mode, comment target), deploy
- **Loot stash** - Collected secrets organized by repository and workflow, with deduplication, source tracking, and one-key pivot recommendations
- **Omnibox search** - Fuzzy search across repos, workflows, jobs, vulnerabilities, and loot items
- **Tab completion** - Phase-aware, context-sensitive suggestions for commands, targets, and arguments
- **Activity log** - Scrollable, timestamped event log with icons for success/warning/error/agent events
- **Help overlay** - `?` or `help` shows all available commands for the current phase
- **Callbacks modal** - Live view of all agent sessions with online/offline status

Teamserver (Kitchen)

- **SSH challenge-response auth** - Operators authenticate with SSH keys via a challenge/verify handshake. The Kitchen sends a nonce, the Counter signs it with the operator's SSH key.
- **Shared-token auth** - Quickstart and E2E environments use a pre-shared `AUTH_TOKEN` for simplified setup
- **BBolt persistence** - Sessions, attack graphs, operation history, known entities, and loot survive Kitchen restarts
- **NATS JetStream** - Durable message bus between Kitchen and Brisket agents. Subjects: `smokedmeat.orders.<agent_id>`, `smokedmeat.coleslaw.<agent_id>`, `smokedmeat.beacon.<agent_id>`
- **GitHub API proxy** - All GitHub API calls are proxied through Kitchen. Counter makes zero direct GitHub API calls. This keeps tokens server-side.

- **Auto-TLS** - Caddy reverse proxy with automatic Let's Encrypt certificates for self-hosted deployments
- **Stager registration** - Kitchen generates unique stager URLs (`/r/{stagerID}`) that serve the Brisket binary on first callback
- **Operation history** - Tracks analysis runs, deployments, and pivot operations with timestamps
- **Live graph endpoint** - `/graph` serves a Cytoscape.js visualization page; `/graph/ws` pushes real-time updates via WebSocket

`docs/WHOOOI.md`

Blob `90abc303d609`, 11582 bytes, at commit `3b2873dbecac`.

Whooli - CI/CD Attack Playground

`whooli` is SmokedMeat's end-to-end CI/CD attack playground. It is a deliberately vulnerable GitHub organization designed to exercise real product workflows: public footholds, runner secret recovery, GitHub pivots, private-repo analysis, cache poisoning, OIDC federation, and cloud post-exploit work.

This document is the canonical guide for the Whooli goat environment. It merges the old challenge guide and the old architecture note into one source of truth.

The Scenario

Whooli is a Silicon Valley giant led by Galvin Belsin. Publicly, the company insists that `xyz` is its own moonshot compression breakthrough and that NewCleus was built independently in-house.

Internally, the story is messier. After failing to reproduce Ritcherd Hendricks's middle-out compression, Whooli rage-acqui-hired Nelson Bigetti from Nip Alert and started scavenging old repo clones, notes, and deployment material from his laptop backups. That is why the private repos read like a panicked reverse-engineering effort and still contain stray references to Danesh and Jilfoyle.

The Objective

Read `flag.txt` from the GCP Cloud Storage benchmark bucket. The flag looks like `SM{...}`.

The shortest validated path starts from a public repo, pivots through a GitHub App key, compromises a private workflow through cache poisoning, federates into GCP with OIDC, and reads the flag from:

- `gs://whooli-newcleus-benchmarks/flag.txt`

Getting Started

```
make quickstart
```

When the setup wizard asks for a target, enter `whooli`.

A classic GitHub PAT with `public_repo` scope is enough to start the main public path. You do not need private-repo access up front because the environment is meant to make you earn it.

Prefer a classic PAT. Fine-grained PATs can be too restrictive and may block public cross-org testing, including `whooli`.

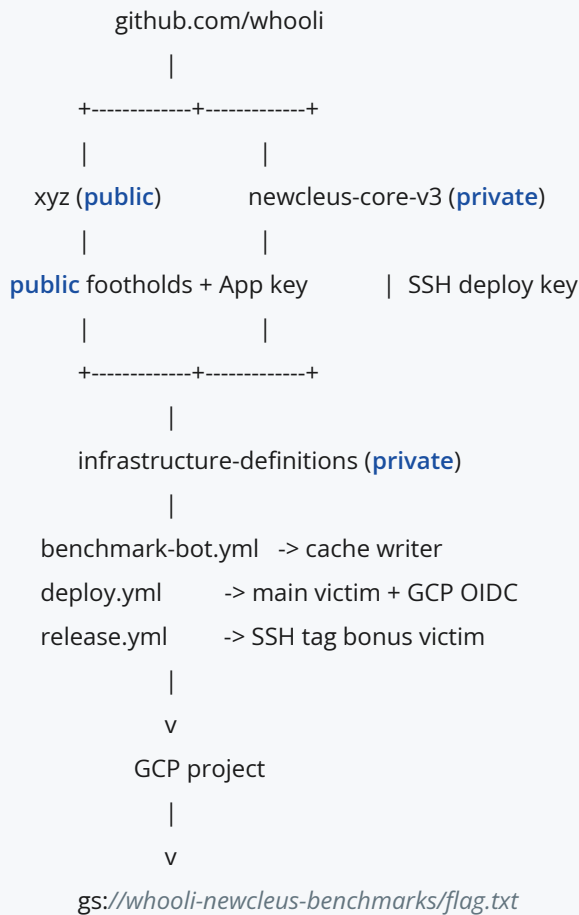
If you are working on SmokedMeat itself, `make dev-quickstart` uses the local source tree instead.

Organization Layout

The `whooli` org contains three repositories:

Repository	Visibility	Role
<code>xyz</code>	Public	Initial foothold and GitHub App-key recovery
<code>infrastructure-definitions</code>	Private	Main cache-poison target and GCP pivot repo
<code>newcleus-core-v3</code>	Private	Alternate challenge path containing the SSH deploy key

Start at <https://github.com/whooli/xyz>. The README there hints at how Whooli's public automation is wired into the private NewCleus effort.



Hard-Validated Main Path

The path below is the one currently hard-validated by `make e2e-goat`. If you want the exact chain SmokedMeat proves end to end today, this is it.

1. Analyze `whooli` and foothold `xyz`

SmokedMeat starts with the public `xyz` repo and surfaces the vulnerable public workflows. The current goat E2E uses an issue-body foothold on:

- `xyz/.github/workflows/auto-labeler.yml`

That gets you runner execution and a first callback.

2. Take the App-key foothold

From Recon, the current goat E2E takes a second public foothold through a comment-driven path that lands in:

- `xyz/.github/workflows/whooli-analyzer.yml`

That second foothold is where the test expects to recover:

- `WHOOOI_BOT_APP_PRIVATE_KEY`

Export the key from the loot stash, then use `pivot app` to mint an installation token. That App token is the main bridge into the private repos because it has the repo visibility and `actions:write` needed for the next phase.

3. Analyze `infrastructure-definitions`

Switch the active token to the installation token, then retarget:

- `whooli/infrastructure-definitions`

SmokedMeat analyzes the private repo and surfaces the writer workflow used in the validated chain:

- `infrastructure-definitions/.github/workflows/benchmark-bot.yml`

This is the current cache-writer foothold used by `make e2e-goat`.

4. Poison the victim cache

Open the exploit wizard for the writer vuln and enable `Cache Poisoning`. In the validated path, the victim is:

- `infrastructure-definitions/.github/workflows/deploy.yml`

The writer callback is immediate. Brisket uses the runner's cache-capable runtime token to stage the poisoned cache entry without needing to modify repo history.

5. Arm dwell, then trigger `deploy.yml`

Victim implants are persistent and default to express mode. Before the victim workflow runs, arm the next implant with dwell so the callback stays available after the poisoned cache restores.

The validated path then triggers:

- `workflow_dispatch` on `deploy.yml`

using the GitHub App installation token. This is the shortest stable path because it stays on the App-token leg and does not require reviving the alternate PAT branch.

6. Pivot to GCP and read the flag

Once the poisoned victim callback lands, the current goat E2E does:

1. `pivot gcp`
2. `cloud shell`
3. `gsutil ls gs://whooli-newcleus-benchmarks/`
4. `gsutil cat gs://whooli-newcleus-benchmarks/flag.txt`

If you can do that chain manually, you have reproduced the validated goat path.

Workflow Map

These are the important workflows in the current environment.

xyz/.github/workflows/auto-labeler.yml

- Trigger: public issue and comment activity
- Role: first public foothold
- Validation status: part of the hard-validated goat path
- Key property: direct shell injection from attacker-controlled issue data

xyz/.github/workflows/whooli-analyzer.yml

- Trigger: public comment-driven path
- Role: second foothold used to recover the GitHub App key
- Validation status: part of the hard-validated goat path
- Key property: lands the runner context where `WHOOOLI_BOT_APP_PRIVATE_KEY` is recoverable

xyz/.github/workflows/internal-sync.yml

- Trigger: `workflow_dispatch`
- Role: alternate bridge into the private repos through `WHOOOLI_INT_PAT`
- Validation status: described challenge path, not part of the hard-validated main goat chain

infrastructure-definitions/.github/workflows/benchmark-bot.yml

- Trigger: `issue_comment`
- Role: main cache writer in the validated path
- Validation status: part of the hard-validated goat path
- Key property: issue-comment injection in a default-branch workflow

infrastructure-definitions/.github/workflows/deploy.yml

- Trigger: `push` on `main` , `workflow_dispatch`
- Role: main victim workflow and GCP OIDC bridge
- Validation status: part of the hard-validated goat path
- Key properties:
 - uses `actions/setup-go@v5` with dependency caching
 - reaches the `actions/checkout` post-run gadget after cache restore
 - has `id-token: write`
 - runs under the hardened workload identity policy

infrastructure-definitions/.github/workflows/release.yml

- Trigger: tag push such as `bench-v*`
- Role: bonus SSH-tag victim path
- Validation status: alternate path, not part of the hard-validated main goat chain

Why the Main Path Works

The important design point is that Whooli does not need a PR-based writer for the main path. A default-branch issue or comment workflow is already enough to act as a privileged cache writer.

The writer foothold stages a poisoned `setup-go` cache entry. The victim `deploy.yml` run is legitimate and still satisfies the OIDC identity checks. What changes is the cached data restored into that legitimate workflow context, not the workflow file itself.

That is why the hardened workload identity provider does not save the environment from the validated attack. The workload identity policy is strict, but it is evaluating a valid workflow identity that is running compromised restored content.

OIDC Hardening

The Whooli GCP workload identity provider is intentionally locked to `deploy.yml` on `main` :

```
assertion.job_workflow_ref == 'whooli/infrastructure-definitions/.github/workflows/deploy.yml@refs/heads/main'
&& assertion.ref == 'refs/heads/main'
&& assertion.ref_protected == 'true'
&& assertion.runner_environment == 'github-hosted'
```

This matters because it explains why a direct off-branch runner pivot is not enough, and why the validated goat path uses cache poisoning to inherit the trusted victim workflow's identity.

Alternate Paths

These routes are part of the challenge design and are still meaningful, but they are not the currently hard-validated `make e2e-goat` chain.

Alternate PAT Path

`xyz/.github/workflows/internal-sync.yml` can still expose:

- `WHOOOI_INT_PAT`

That PAT gives an alternate route into the private repos. It is useful challenge material, but the shortest validated path does not need it.

Alternate SSH Path

`newcleus-core-v3` still contains the SSH deploy key for `infrastructure-definitions` .

The intended bonus path is:

1. extract the deploy key from `newcleus-core-v3`
2. poison the `release.yml` cache from the main writer foothold
3. push a tag such as `bench-v2026.03.24` over SSH

4. let `release.yml` restore the poisoned cache and execute in the tagged-release context

This is still described here because it is part of the environment design, even though the main E2E goat test does not use it.

Practical Operator Notes

- Run `analyze` first. The Whooli environment is designed to be discovered through the product, not by memorizing file names.
- Check the loot stash after each callback. The important pivot material changes as you move from public foothold to private infra.
- Use the wizard for cache poisoning. The product already handles writer-victim pairing, cache prediction, implant staging, and victim arming.
- If `pivot gcp` fails, confirm you are in the `deploy.yml` victim callback, not just in the writer callback.

Defensive Lessons

1. Never interpolate issue or comment content directly into `run:` blocks.
2. Treat default-branch issue and comment workflows as privileged cache writers.
3. Do not assume OIDC hardening protects against poisoned restored cache content.
4. Keep GitHub App scope minimal across repositories.
5. Treat SSH deploy keys as high-value credentials even when they are no longer part of the shortest attack path.

Rules of Engagement

- `whooli` is a shared test environment. Issues and PRs created during your run are automatically cleaned up by the E2E harness where possible.
- If you are running manually, clean up after yourself and close anything you created.
- The flag rotates. Prove the path, not the flag value.
- The hard-validated path above is authoritative for product regression checking. The alternate paths are intentionally left as challenge material and may evolve independently.

`docs/adr/0001-separate-kitchen-schema-version-from-application-releases.md`

Blob `7cb3c9076d3e`, 628 bytes, at commit `3b2873dbecac`.

Separate Kitchen schema versions from application releases

Kitchen persists engagement state across application upgrades. Its database therefore carries an explicit schema version that is independent of the SmokedMeat release version: additive changes may advance the schema minor, while changes that older binaries or existing data cannot

interpret safely advance the schema major and make startup fail with clear purge guidance instead of applying an implicit migration. This favors predictable compatibility and honest failure over coupling storage changes to release numbers or accumulating an early migration framework.

`docs/agents/domain.md`

Blob `af93c7a7fb77`, 1225 bytes, at commit `3b2873dbecac`.

Domain Docs

How the engineering skills should consume this repository's domain documentation when exploring the codebase.

Before exploring, read these

- `CONTEXT.md` at the repository root, or
- `CONTEXT-MAP.md` at the repository root if it exists. It points to one `CONTEXT.md` per context.
- `docs/adr/` for decisions affecting the area being changed.

If these files do not exist, proceed silently. Do not flag their absence or suggest creating them upfront. The `/domain-modeling` skill creates them lazily when terminology or decisions are resolved.

File structure

This repository uses the single-context layout:

```
/
  CONTEXT.md
  docs/adr/
    0001-separate-kitchen-schema-version-from-application-releases.md
  internal/
```

Use the glossary's vocabulary

When output names a domain concept, use the term defined in `CONTEXT.md`. Do not drift to synonyms that the glossary explicitly avoids.

If a needed concept is absent, reconsider whether the term belongs to the project or note the gap for `/domain-modeling`.

Flag ADR conflicts

If output contradicts an existing ADR, surface the conflict explicitly instead of silently overriding it.

Blob `fe6f005a828c`, 3384 bytes, at commit `3b2873dbecac`.

Engineering Guide For Agents

Use this guide for repository orientation and verification. Area-specific constraints live in sibling files so agents only load what the task requires.

Working Method

- 1. Read `CONTEXT.md` and relevant ADRs.
- 2. Resolve the ticket and acceptance criteria before editing.
- 3. Inspect the nearest implementation and tests.
- 4. Make the smallest coherent change that satisfies the behavior.
- 5. Verify the narrow behavior first, then widen checks according to risk.
- 6. Review the final diff for unrelated changes and stale documentation.

Preserve unrelated worktree changes. Do not mix roadmap cleanup, refactors, and product behavior in one change unless they are inseparable.

Repository Map

Domain	Location
Counter operator client	<code>cmd/counter/</code> , <code>internal/counter/</code> , <code>internal/counter/tui/</code>
Kitchen teamserver	<code>cmd/kitchen/</code> , <code>internal/kitchen/</code>
Brisket implant	<code>cmd/brisket/</code> , <code>internal/brisket/</code>
Shared protocol models	<code>internal/models/</code>
Pantry attack graph	<code>internal/pantry/</code>
NATS messaging	<code>internal/pass/</code>
Injection payloads and stagers	<code>internal/rye/</code>
LOTP catalog and payloads	<code>internal/lotp/</code>
Poutine analysis integration	<code>internal/poutine/</code>
Secret scanning	<code>internal/gitleaks/</code> , <code>internal/gump/</code>
Integration and operator E2E	<code>tests/</code> , <code>.claude/e2e/</code>

Commands

Run `make help` for the complete current command list.

Intent	Command
Focused package test	<code>go test ./path/to/package</code>
Full unit suite	<code>make test</code>
Lint	<code>make lint</code>
Integration tests	<code>go test -tags=integration ./...</code>
Public exploit smoke path	<code>make e2e-smoke</code>
Full Whooli chain	<code>make e2e-goat</code>
Local development stack	<code>make dev-quickstart</code>
Stable release quickstart	<code>make quickstart</code>
Rebuild Kitchen during manual E2E	<code>make e2e-kitchen-rebuild</code>
Build Brisket	<code>make build-brisket</code>

Use `make dev-quickstart-purge` or `make quickstart-purge` only when state is intentionally disposable or a schema-major mismatch requires it.

Test Selection

- Counter state or rendering changes: focused TUI tests, then `go test ./internal/counter/tui`.
- Kitchen handler or protocol changes: focused handler tests and affected client tests.
- Persisted data changes: DB schema tests plus Kitchen package tests.
- Brisket behavior: platform-appropriate Brisket tests; use integration coverage when process or runner behavior matters.
- Cross-process flows: the smallest relevant E2E target.
- Documentation-only changes: link inspection, `git diff --check`, and any generator check affected by the docs.

Tests should prove user-visible behavior, security boundaries, state transitions, and regressions. Avoid tests whose only purpose is increasing coverage.

Generated And Release-Sensitive Files

- After changing GitHub Actions workflows, run `make pinact`.
- Update `configs/quickstart-release.mk` only through `make quickstart-pin VERSION=v...`.
- Use `make tag VERSION=v...` only for an intentional signed release.
- Keep `CHANGELOG.md` aligned with published GitHub releases and meaningful unreleased changes.

Worktrees

For parallel feature work, follow `docs/worktrees-flow.md` . Keep one coherent feature per branch and avoid stacking unless the dependency is real.

`docs/agents/issue-tracker.md`

Blob `a91963af6e98` , 3198 bytes, at commit `3b2873dbecac` .

Work Tracking: GitHub Issues and GitHub Projects

Public issues for this repository live in GitHub Issues. Maintainer-owned roadmap ideas live as draft items in the private SmokedMeat Roadmap Project.

Use the `gh` CLI for all GitHub operations.

Destinations

GitHub Issues

Use GitHub Issues for:

- community bug reports and feature requests
- security, correctness, and usability defects
- work that is ready for public discussion
- implementation-ready tickets
- work that contributors can claim

SmokedMeat Roadmap

Maintainer-owned roadmap ideas live in:

- Owner: `boostsecurityio`
- Project number: `4`
- Project title: `SmokedMeat Roadmap`
- URL: `https://github.com/orgs/boostsecurityio/projects/4`

Use Project draft items for ideas that should not appear in the repository's public Issues list.

Draft items start with Status `Backlog` . Leave Priority and Size unset until the item has been reviewed.

Convert a draft item to a repository issue when it is ready for public discussion, implementation, or contributor ownership.

GitHub Issue conventions

- **Create an issue:** `gh issue create --title "..." --body "..."`
- **Read an issue:** `gh issue view <number> --comments`
- **List issues:** `gh issue list --state open --json number,title,body,labels,comments`

- **Comment:** `gh issue comment <number> --body "..."`
- **Apply or remove labels:** `gh issue edit <number> --add-label "..."` or `--remove-label "..."`
- **Close:** `gh issue close <number> --comment "..."`

Infer the repository from `git remote -v`.

Project conventions

- **List items:** `gh project item-list 4 --owner boostsecurityio`
- **Create a roadmap draft:** `gh project item-create 4 --owner boostsecurityio --title "... --body "..."`
- **Add an existing issue:** `gh project item-add 4 --owner boostsecurityio --url "<issue-url>"`
- **Inspect fields:** `gh project field-list 4 --owner boostsecurityio`

The Project Status values are:

- Backlog
- Ready
- In progress
- In review
- Done

The Project Priority values are `P0`, `P1`, and `P2`.

The Project Size values are `XS`, `S`, `M`, `L`, and `XL`.

Routing skill output

When a skill says "publish to the issue tracker":

- Create a Project draft item when the output is a maintainer-owned roadmap idea, exploratory proposal, or uncommitted future direction.
- Create a GitHub issue when the work is implementation-ready, intended for public discussion, or explicitly requested as an issue.
- Add newly created implementation issues to the SmokedMeat Roadmap Project.

When a skill says "fetch the relevant ticket":

- Use `gh issue view` for a GitHub issue.
- Use `gh project item-list` and the Project APIs for a draft item.

Pull requests as a triage surface

PRs as a request surface: no.

Pull requests are implementation artifacts, not substitutes for issue or roadmap intake.

Planning documentation

GitHub Issues and the SmokedMeat Roadmap Project are the sources of truth for work status.

Do not create a replacement `docs/ROADMAP.md` or new planning files under `docs/tasks/`.

Durable product behavior belongs in product documentation. Durable architectural decisions belong in `docs/adr/`.

`docs/agents/persistence.md`

Blob `3135cfc2fcef`, 1955 bytes, at commit `3b2873dbecac`.

Kitchen Persistence Guide

Read this file before changing `internal/kitchen/db/` or any persisted Pantry, stager, session, loot, history, or known-entity shape.

Kitchen schema versions are independent of application release versions. See ADR-0001.

Compatibility Invariant

Within one schema major:

- newer binaries must open older data with safe defaults
- older binaries must not misinterpret data written by newer binaries
- patch and minor application releases must preserve existing quickstart and development volumes

If either direction is unsafe, advance the schema major and fail fast with guidance to run `make quickstart-purge` or `make dev-quickstart-purge`.

Version Decisions

Advance the schema minor for additive, backward-compatible disk changes:

- new buckets that do not change existing meaning
- optional persisted fields
- optional Pantry properties or relationships
- metadata or derived state that can be defaulted or rebuilt

Do not change the schema version for:

- Counter-only behavior
- in-memory changes that preserve serialized shape and meaning
- tests, documentation, logging, or internal refactors

Advance the schema major for incompatible changes:

- renamed, removed, or repurposed buckets
- changed key layout or identity rules
- newly required fields

- changed meaning of persisted values
- serialization changes that existing data cannot be read safely
- changes that require a one-off migration or operator purge

Legacy Data

When a database predates schema metadata but contains the known current buckets, treat it as legacy-current and backfill schema metadata on open.

Verification

At minimum, run:

```
go test ./internal/kitchen/db ./internal/kitchen
```

Include compatibility coverage for legacy-current data, safe minor evolution, and schema-major mismatch behavior whenever the persisted contract changes.

docs/agents/triage-labels.md

Blob `f843b5ebc976`, 1035 bytes, at commit `3b2873dbecac`.

Triage Labels

The skills use five canonical triage roles. This file maps those roles to the GitHub labels used by this repository.

Label in mattpocock/skills	GitHub label	Meaning
<code>needs-triage</code>	<code>needs-triage</code>	Maintainer needs to evaluate this issue
<code>needs-info</code>	<code>needs-info</code>	Waiting on reporter for more information
<code>ready-for-agent</code>	<code>ready-for-agent</code>	Fully specified, ready for an AFK agent
<code>ready-for-human</code>	<code>ready-for-human</code>	Requires human implementation
<code>wontfix</code>	<code>wontfix</code>	Will not be actioned

When a skill mentions a triage role, use the corresponding GitHub label.

Project Status and triage labels serve different purposes. Status tracks execution progress, while labels record the triage outcome and suitable implementation owner.

docs/agents/tui.md

Blob `3f093826128c`, 1967 bytes, at commit `3b2873dbecac`.

Counter TUI Guide

Read this file before changing `internal/counter/tui/` .

Ownership

Keep `update.go` as the main Bubble Tea dispatcher. Put behavior in the subject file that owns it:

Subject	Files
Setup and authentication	<code>setup.go</code> , <code>token.go</code>
Exploit wizard and delivery	<code>wizard.go</code> , <code>deploy.go</code>
Commands and suggestions	<code>command.go</code> , <code>suggestions.go</code>
Kitchen events and agents	<code>kitchen.go</code> , <code>agent.go</code>
Analysis and pivots	<code>analysis.go</code> , <code>pivot.go</code>
Tree and rendering	<code>tree.go</code> , <code>view.go</code> , <code>layout.go</code>
State and messages	<code>model.go</code> , <code>phase.go</code> , <code>messages.go</code>

Do not grow `update.go` with subject-specific logic.

Layout Invariants

- Use `ultraviolet/layout` splits with `layout.Percent` or `layout.Fixed` . Do not calculate panel sizes with manual width or height arithmetic.
- Compose overlays with `uv.ScreenBuffer` : draw the background, then draw the modal.
- Do not use `Lipgloss Height()` . Pad content explicitly before rendering.
- Use `Lipgloss` borders. Do not hand-draw Unicode box borders.
- Pad modal content lines to the full inner width so the right border remains aligned.
- Route terminal dimensions through `image.Rectangle` areas and existing layout helpers.

These are correctness constraints, not style preferences. Violating them causes initial-render duplication, ANSI corruption, or font-dependent alignment failures.

Interaction And Tests

- Keep state transitions independent of rendering where practical.
- Test keyboard transitions, escape paths, invalid input, reconnect behavior, and small terminal sizes.
- Use table-driven tests for behavior variants.
- Use golden or render tests only when visual structure is the behavior under test.
- Set a stable color profile for output assertions.

Run:

```
go test ./internal/counter/tui
```

Use `make e2e-smoke` or `make e2e-goat` when the change crosses Counter, Kitchen, delivery, or shell boundaries.

`docs/deployment.md`

Blob `8b50159f5deb`, 3465 bytes, at commit `3b2873dbecac`.

SmokedMeat Deployment Guide

Modes

Development Quickstart

Use this when you want the fastest local path:

```
make dev-quickstart
```

Development quickstart uses:

- Docker for cloudflared, NATS, and Kitchen
- a Cloudflare tunnel for a public Kitchen URL
- shared-token auth (`AUTH_MODE=token`)
- local `go run ./cmd/counter`
- the `smokedmeat-cloud-shell` image for `cloud shell` and `ssh shell`

This is the right path for demos, local testing, and validating the full workflow without standing up a domain.

`make quickstart` uses the pinned release in `configs/quickstart-release.mk`. That pin can intentionally lag the newest tag until a release has been validated. Use `make quickstart-version` to inspect it. Maintainers update it with `make quickstart-pin VERSION=v...`, which verifies the immutable GitHub release plus the signed GHCR image digests before writing the pin. `make quickstart-pin` requires `gh`, `cosign`, and `docker` in `PATH`. Use `make quickstart-up` / `make quickstart-counter` if you want to split infrastructure startup from the Counter launch.

Counter checks for newer SmokedMeat releases at startup. It can be disabled by setting the `SMOKEDMEAT_DISABLE_VERSION_CHECK` environment variable.

Self-Hosted Kitchen

Use this when you want a stable Kitchen with your own domain and SSH challenge-response auth.

Prerequisites

- Docker and Docker Compose on the Kitchen host

- a DNS record pointing your Kitchen hostname to that host
- an SSH agent on the operator workstation
- this repository checked out on the Kitchen host

1. Register an operator key

From the operator workstation, list keys from the local SSH agent:

```
go run ./cmd/counter --list-keys
```

Pick one entry and copy the printed `authorized_keys` line.

On the Kitchen host, add that line to:

```
~/.smokedmeat/authorized_keys
```

Kitchen reads that file from the host and mounts it into the container.

2. Start Kitchen

From the repo root on the Kitchen host:

```
export DOMAIN=kitchen.example.com
docker compose -f deployments/docker-compose.yml up -d --build
```

This stack starts:

- Caddy on ports `80` and `443`
- Kitchen behind Caddy
- NATS JetStream
- BBolt persistence under the Docker volume

By default this path uses:

- `AUTH_MODE=ssh`
- automatic TLS via Caddy
- `~/.smokedmeat/authorized_keys` for operator auth

3. Connect Counter

From the operator workstation:

```
go run ./cmd/counter -kitchen https://kitchen.example.com -operator <name>
```

If the Kitchen URL and operator name are already stored in `~/.smokedmeat/config.yaml`, `make counter` is enough:

```
make counter
```

Counter will request a challenge from Kitchen and sign it with the SSH key in the local agent.

4. Cloud Shell Support

If Counter runs as a local Go binary on the operator workstation, build the cloud shell image once:

```
make cloud-shell-image
```

`make dev-quickstart` already builds this image for you. `make quickstart` uses the pinned released image that matches the downloaded Counter binary.

Local Development Compose

If you want Kitchen exposed directly on localhost instead of going through Caddy:

```
docker compose -f deployments/docker-compose.yml -f deployments/docker-compose.dev.yml up -d --build
```

That exposes:

- Kitchen on `http://localhost:8080`
- NATS on `localhost:4222`

Auth still defaults to SSH mode unless you explicitly override `AUTH_MODE`.

docs/worktrees-flow.md

Blob 13abf42ca511, 2036 bytes, at commit 3b2873dbecac.

Worktrees Flow

Use Git worktrees when you want multiple Codex sessions against the same repo without stashing or branch hopping.

Default Pattern

Keep the primary checkout on `main`. Put feature worktrees under `../smokedmeat-wt/`. Cut every independent feature branch from the same fresh `origin/main`.

```
make worktree-sync-main
make worktree-add NAME=session-a
make worktree-add NAME=session-b
make worktree-add NAME=session-c
make worktree-list
```

Defaults:

- `make worktree-add NAME=foo` creates branch `feat/foo`
- the worktree path is `../smokedmeat-wt/foo`
- the base is `origin/main`

Open one Codex session per worktree. Keep the original checkout on `main` as the clean control workspace.

When To Override Defaults

Use a custom branch name when the default `feat/<name>` does not fit:

```
make worktree-add NAME=auth-cleanup BRANCH=fix/auth-cleanup
```

Use a custom destination when you want a different worktree root:

```
make worktree-add NAME=api-cleanup DEST=../alt-worktrees/api-cleanup
```

Use a non-`main` base only for stacked work that truly depends on another open branch:

```
make worktree-add NAME=followup BRANCH=feat/followup BASE=feat/base
```

Review Flow

- Keep one feature per worktree and one PR per branch.
- For unrelated work, branch every PR directly from `origin/main`.
- Open PRs as soon as the branch is coherent, but keep them small enough to review independently.
- Add fixup commits during review. Do not rebase after every comment.
- Rebase or merge `origin/main` only when the branch is next to land, has a real conflict, or CI requires an up-to-date base.
- Use stacked branches only when one change truly depends on another. Otherwise every rewrite of the base branch creates more rebase work.

Cleanup

After a branch lands:

```
make worktree-remove NAME=session-a  
make worktree-prune
```

`worktree-remove` removes the worktree directory, then tries `git branch -d` . If the branch is not merged yet, Git keeps the branch and prints a note.

Archived from <https://github.com/boostsecurityio/smokedmeat>. Rendered offline from the local Markdown copy.